

ZIGBEE AND LOW-POWER WIRELESS NETWORKS: ARCHITECTURE, APPLICATIONS, AND CHALLENGES

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INTRODUCTION

- Rapid growth of the Internet of Things (IoT)
- Need for energy-efficient wireless communication
- Wi-Fi limitations for battery-powered devices
- ZigBee as a low-power wireless alternative
- Objective: Explore ZigBee architecture, applications & challenges



LOW-POWER WIRELESS NETWORKS

Characteristics

- Low power consumption
- Long battery life
- Low implementation cost
- Mesh networking capability
- Scalability for large deployments



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Applications

- Smart homes
- Healthcare monitoring
- Environmental monitoring
- Industrial automation
- Wireless Sensor Networks (WSNs)



[3]

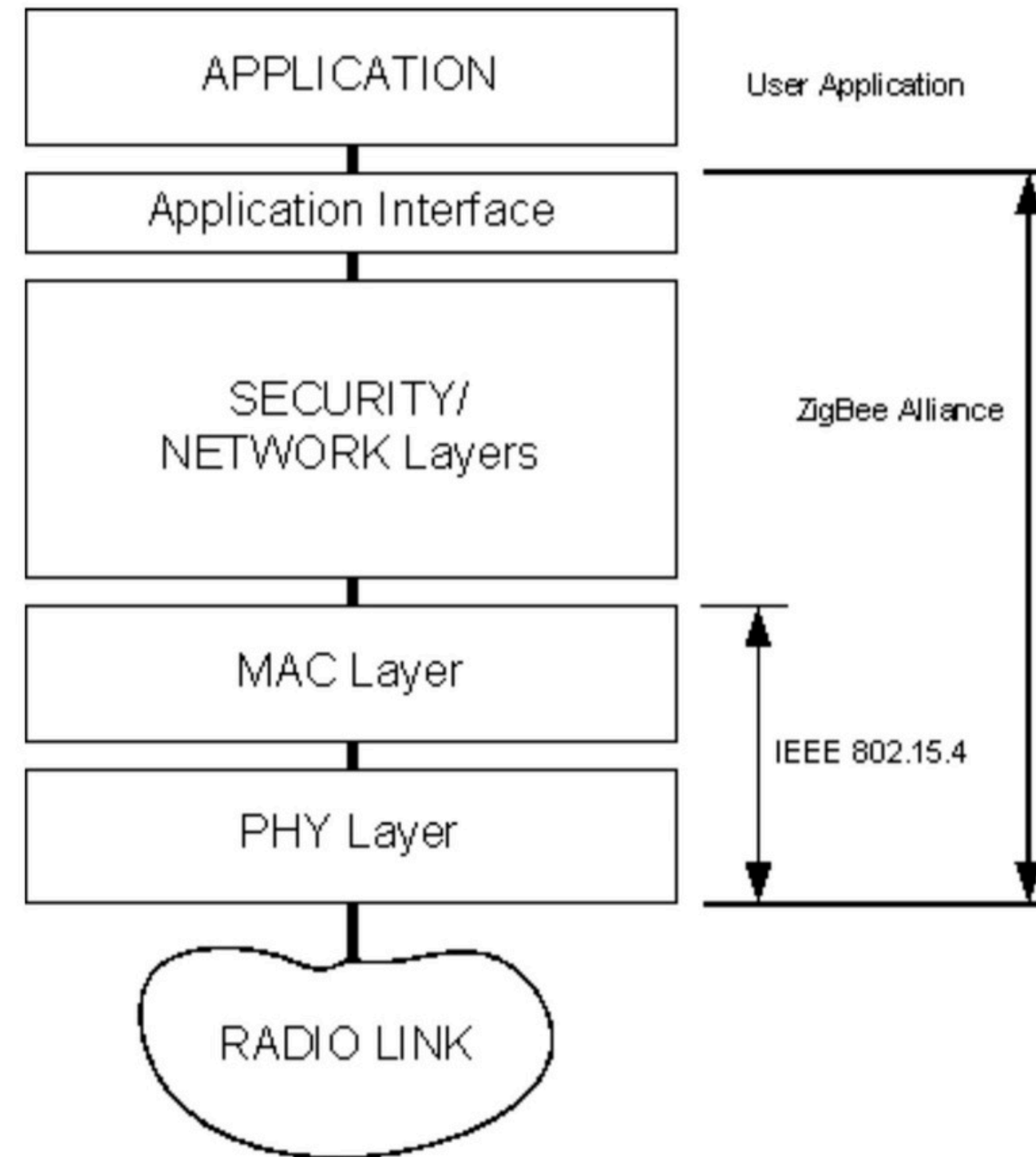
ZIGBEE ARCHITECTURE & IEEE 802.15.4

Protocol Layers:

- Based on IEEE 802.15.4 standard
- Physical (PHY) Layer
- MAC Layer
- Network (NWK) Layer
- Application (APL) Layer

Key Features:

- Mesh networking
- Routing capabilities
- Security mechanisms
- Device interoperability



NETWORK TOPOLOGIES

Topology Types:

- Star Topology – centralized communication
- Tree Topology – hierarchical, extended range
- Mesh Topology – decentralized, multi-path

Key Highlights:

- Mesh enables self-healing networks
- Reliable, redundant communication
- Highly scalable for dense deployments

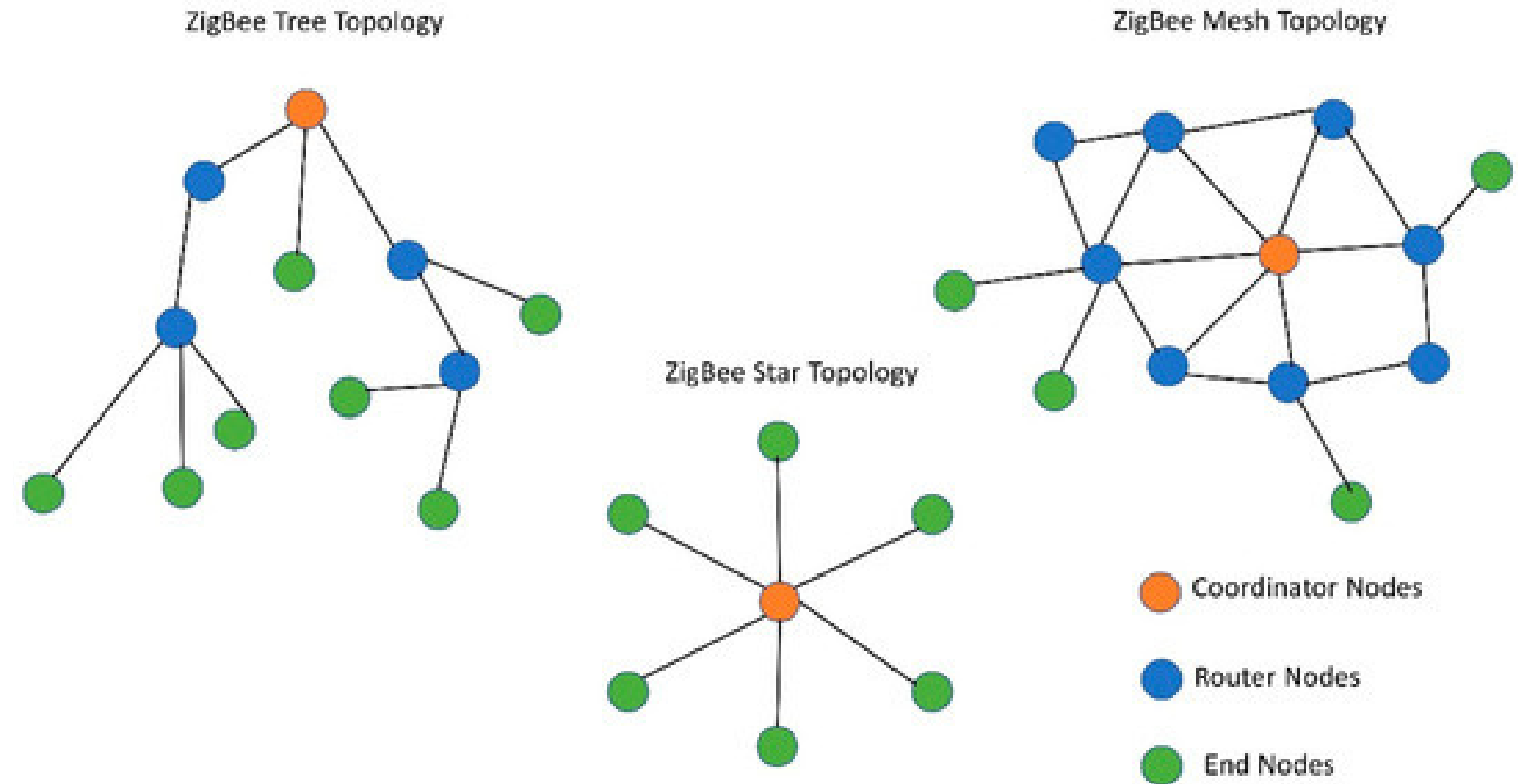


Figure: Star, Tree & Mesh Topologies

INDUSTRIAL APPLICATIONS

Smart Factory Applications:

- Temperature Monitoring – prevents overheating
- Humidity Monitoring – protects sensitive equipment
- Real-Time Monitoring – live sensor data

Key Benefits:

- Predictive maintenance
- Lower maintenance costs
- Improved machine safety



Figure: ZigBee-enabled Smart Factory Monitoring

SECURITY

Security Features

- AES-128 Encryption for data confidentiality
- Authentication protocols to verify devices
- Data integrity checks to prevent tampering
- Trust Center managing and distributing security keys
- Network-layer and application-layer security



Challenges & Solutions

Challenges:

- Vulnerability to replay attacks
- Risk of eavesdropping
- Susceptibility to DoS attacks

Solutions:

- Regular firmware updates
- Robust key management procedures
- Strict access control mechanisms



COMPARISON: ZigBee vs BLE vs LoRaWAN



ZigBee

- Medium Range
- Low Power
- Mesh Networking
- Industrial IoT
- Long Battery Life



Bluetooth Low Energy (BLE)

- Short Range
- Very Low Power
- High Data Rate
- Smartphone Compatible
- Wearable Devices



LoRaWAN

- Long Range
- Very Low Power
- Low Data Rate
- Smart Agriculture
- Smart Cities

ENERGY HARVESTING



Solar Energy

- Uses photovoltaic panels
- Powers outdoor ZigBee sensors
- Reduces battery replacement



RF Energy Harvesting

- Harvests ambient RF signals
- Supports ultra-low-power sensors
- Supplementary energy source



Mechanical Vibration

- Converts vibration into electricity
- Suitable for industrial monitoring
- Enables maintenance-free operation

Key Benefits

- Longer Battery Life
- Reduced Maintenance
- Self-Powered IoT

CHALLENGES & FUTURE TRENDS

Challenges

- Interference from other wireless systems (Wi-Fi, Bluetooth)
- Scalability limitations in ultra-large deployments
- Energy consumption of routers in mesh networks
- Limited data rate compared to modern standards
- Security vulnerabilities in dense IoT environments



Future Trends

- Integration with Industry 4.0 initiatives
- Edge computing for local data processing
- Cloud integration for analytics and management
- Adoption of the Matter standard for interoperability
- Development of smart infrastructure ecosystems



CONCLUSION

Main Findings

- Low-power and energy-efficient communication
- Reliable mesh networking
- Secure communication using AES-128
- Suitable for Industrial IoT and Wireless Sensor Networks
- Future-ready with energy harvesting and smart infrastructure



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THANK YOU

Any Questions?

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